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(Portfolio)

Top Skills

Research Technology

Scholarly Research

Metabolomics

Publications

OPTIMIZATION OF DIFFERENT PARAMETERS FOR CULTIVATION OF SPINACIA OLERACEA IN NFT HYDROPONICS

Conservation of Medicinal Plant Bramhi- *Bacopa monnieri*(L.) Wettstein Through in vitro Cultures

In vitro Strategies for the Production of Bioactive Therapeutics from *Caralluma* Species

Effect of Jasmonic Acid on Somatic Embryogenesis in *Caralluma fimbriata*

Efficient In vitro Propagation and Optimized Multiple Shoot Induction of *Caralluma bhupenderiana* an Endangered Medicinal Plant

Dr. Jaswanth Boddupalli Ph. D.

Postdoctoral Researcher (Ph.D. Biotechnology) | Plant Metabolomics & Natural Product Drug Discovery | GC-MS | Molecular Docking & Dynamics | IISc Bengaluru
Greater Bengaluru Area

Summary

I am a postdoctoral-level biotechnology researcher at the Indian Institute of Science (IISc), Bengaluru, operating across the full natural product discovery pipeline: endangered medicinal plant conservation, in vitro propagation, phytochemical metabolomics, and in silico therapeutic screening. My doctoral research produced peer-reviewed publications and book chapters on in vitro propagation of endangered medicinal taxa, somatic embryogenesis, elicitor-mediated bioactive production in the genus *Caralluma*, micropropagation of *Bacopa monnieri*, and Nutrient Film Technique (NFT) hydroponics optimization. Core Technical Toolkit: • Plant Metabolomics & Phytochemistry: GC-MS and FT-IR spectral fingerprinting, secondary metabolite extraction, and elicitor-regulated accumulation (jasmonic acid, salicylic acid). • Plant Tissue Culture & Micropropagation: Somatic embryogenesis, multiple shoot induction, callus/suspension culture, and germplasm conservation. • In Silico Drug Discovery: Molecular docking, molecular dynamics (MD) simulations, and ligand-target binding energy characterization. • Controlled-Environment Agriculture (CEA): Nutrient Film Technique (NFT) hydroponics, nutrient solution dynamics (EC/pH monitoring), and growth modeling. • Computational Tooling: Python (numpy, pandas, matplotlib), RDKit, reproducible data pipelines. What I Bring to a Research Group: A rare bench-to-computation workflow: I bridge physical botanical conservation, wet-lab analytical chemistry (GC-MS), and in silico virtual screening within a single reproducible methodology. Current Research Interests: Microfluidics and optical bio-instrumentation, computational metabolomics, and translational natural product drug discovery. Open to postdoctoral fellowships and collaborative research opportunities in metabolomics, drug discovery, and natural products biotechnology globally. Contact: jaswanthkrish96@gmail.com | Bengaluru, India Google Scholar: Dr. Boddupalli Krishna Jaswanth

Experience

Indian Institute of Science (IISc)

Project Associate II

April 2026 - Present (7 months)

Bengaluru

Department: Optics and Microfluidics Instrumentation (OMI) Lab, Dept. of Instrumentation and Applied Physics

- Conduct research at the intersection of optical instrumentation, microfluidic platforms, and biological sample evaluation within the Optics and Microfluidics Instrumentation (OMI) Lab.
- Apply analytical and spectroscopic methodologies to evaluate biomolecular targets and secondary metabolite kinetics.
- Collaborate on cross-disciplinary bio-instrumentation workflows and manuscript preparation.

Vikrama Simhapuri University (VSU), Nellore

Senior Project Fellow

September 2024 - March 2026 (1 year 7 months)

Sri Potti Sriramulu Nellore

- Conducted advanced research on elicitor-mediated secondary metabolite enhancement in endangered *Caralluma* species.
- Executed high-resolution spectroscopic (GC-MS/FT-IR) metabolomic fingerprinting under jasmonic acid and salicylic acid induction regimes.
- Supervised undergraduate and junior postgraduate laboratory researchers in sterile culture techniques, media preparation, and analytical assays.

Vikrama Simhapuri University College

Researcher Ph. D Student

October 2021 - March 2026 (4 years 6 months)

India

- Completed doctoral dissertation focused on in vitro propagation, conservation, and therapeutic evaluation of endangered Indian medicinal taxa.
- Developed optimized protocols for somatic embryogenesis in *Caralluma fimbriata* and multiple shoot induction in *Caralluma bhupenderiana*.
- Evaluated controlled-environment cultivation parameters using Nutrient Film Technique (NFT) hydroponics for *Spinacia oleracea* and *Capsicum annuum*.
- Performed molecular docking and molecular dynamics (MD) simulations to assess the binding potential of plant-derived compounds against disease targets.

Education

Vikrama Simhapuri University (VSU), Nellore

Doctor of Philosophy - PhD, Biotechnology · (October 2021 - May 2026)

Sri Venkateswara University

Postgraduate Degree, Biotechnology- Gold medalist · (June 2018 - January 2021)

Rashtriya Sanskrit Vidyapeetha

P.G. Diploma , Yoga Teacher Training/Yoga Therapy · (August 2018 - March 2019)

Sri Venkateswara University

Bachelor of Applied Science - BASc, Biotechnology · (2015 - 2018)

Viswa sai

Intermediate , Biology, physics, chemistry · (June 2011 - March 2013)